

Week 4.

CREATE A CART AND PLACE ORDER FUNCTION

A. Database

You need to add three tables in BookStore database:

- orders table design:

	Column Name	Data Type	Allow Nulls
PK	order_id	int	☐
	ispayment	bit	☒
	isship	bit	☒
	order_date	date	☒
	delivery_date	date	☒
	customer_id	int	☒

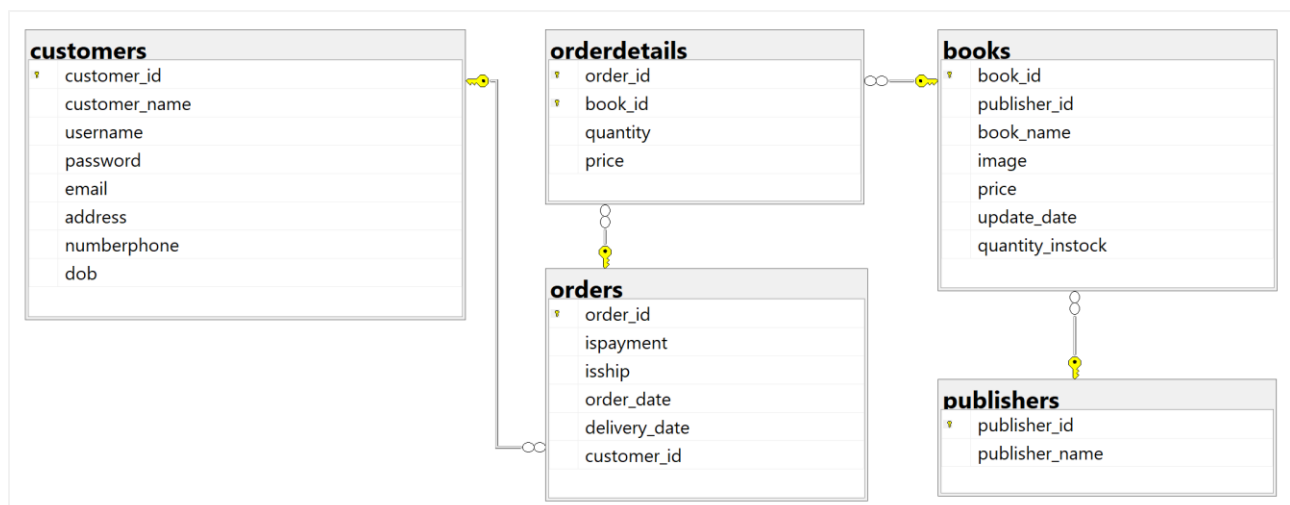
- orderdetails table design:

	Column Name	Data Type	Allow Nulls
PK	order_id	int	☐
FK	book_id	int	☐
	quantity	int	☒
	price	decimal(18, 0)	☒
			☐

- customers table design:

	Column Name	Data Type	Allow Nulls
PK	customer_id	int	☐
	customer_name	nvarchar(50)	☒
	username	varchar(20)	☒
	password	varchar(10)	☒
	email	varchar(30)	☒
	address	nvarchar(100)	☒
	numberphone	varchar(12)	☒
	dob	smalldatetime	☒

- Database diagram:



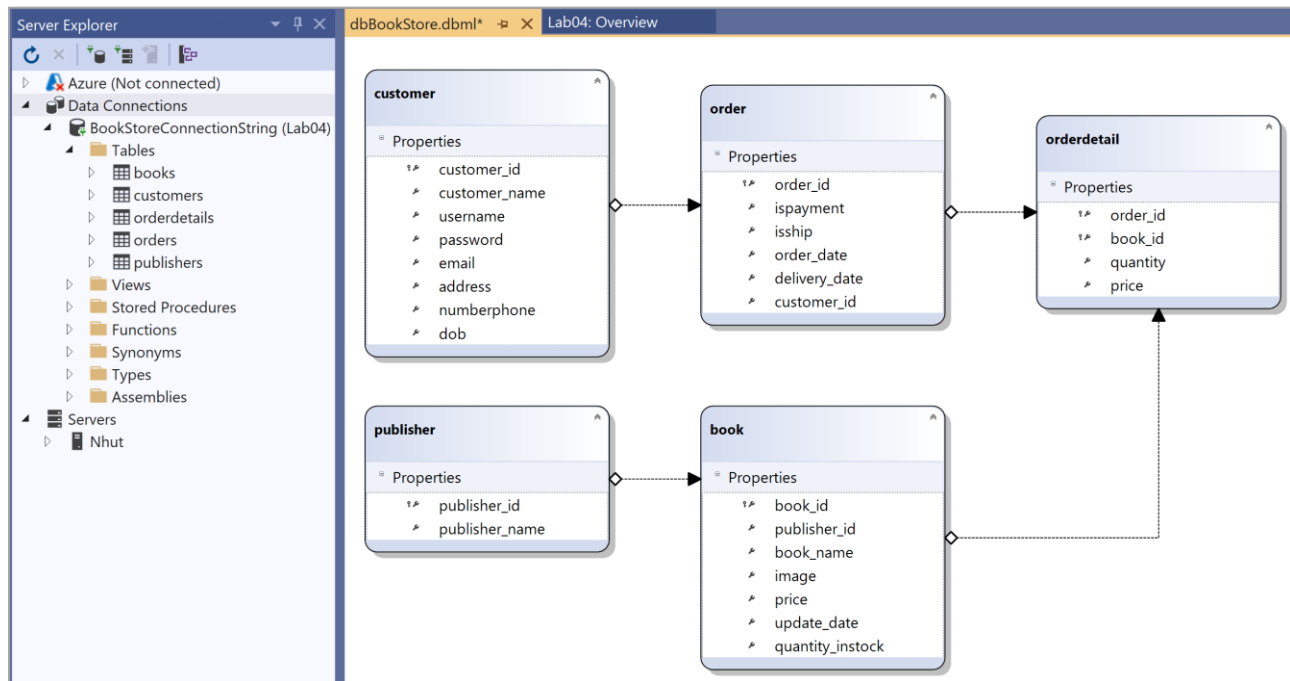
The script to create the tables is:

```
create database BookStore1
go
use BookStore1
go
create table publishers
(
    publisher_id int identity(1,1) primary key not null,
    publisher_name nvarchar(50)
)
go
create table books
(
    book_id int identity(1,1) primary key not null,
    book_name nvarchar(50) not null,
    image varchar(50),
    price decimal(18,0),
    update_date smalldatetime,
    quantity_instock int,
    publisher_id int references publishers(publisher_id)
)
go
create table customers(
    customer_id int identity(1,1) primary key,
    customer_name nvarchar(50),
    username varchar(20),
    password varchar(10),
    email varchar(30),
    address nvarchar(100),
    numberphone varchar(12),
    dob smalldatetime
)
go
create table orders(
    order_id int identity(1,1) primary key,
    ispayment bit,
    isship bit,
    order_date date,
    delivery_date date,
    customer_id int references customers(customer_id)
)
go
create table orderdetails(
    order_id int references orders(order_id),
    book_id int references books(book_id),
    quantity int,
    price decimal(18,0),
    primary key(order_id,book_id)
)
go
insert into publishers(publisher_name) values(N'Publisher 1')
insert into publishers(publisher_name) values(N'Publisher 2')
insert into publishers(publisher_name) values(N'Publisher 3')
insert into publishers(publisher_name) values(N'Publisher 4')
select * from publishers
insert into books(book_name,image,price,update_date,quantity_instock)
values(N'Book 1','Context/images/book1.jpg',3.5,'12/09/2023',20)
insert into books(book_name,image,price,update_date,quantity_instock)
values(N'Book 2','Context/images/book2.jpg',3.5,'12/09/2023',20)
insert into books(book_name,image,price,update_date,quantity_instock)
values(N'Book 3','Context/images/book3.jpg',3.5,'12/09/2023',20)
insert into books(book_name,image,price,update_date,quantity_instock)
values(N'Book 4','Context/images/book4.jpg',3.5,'12/09/2023',20)
select * from books
```

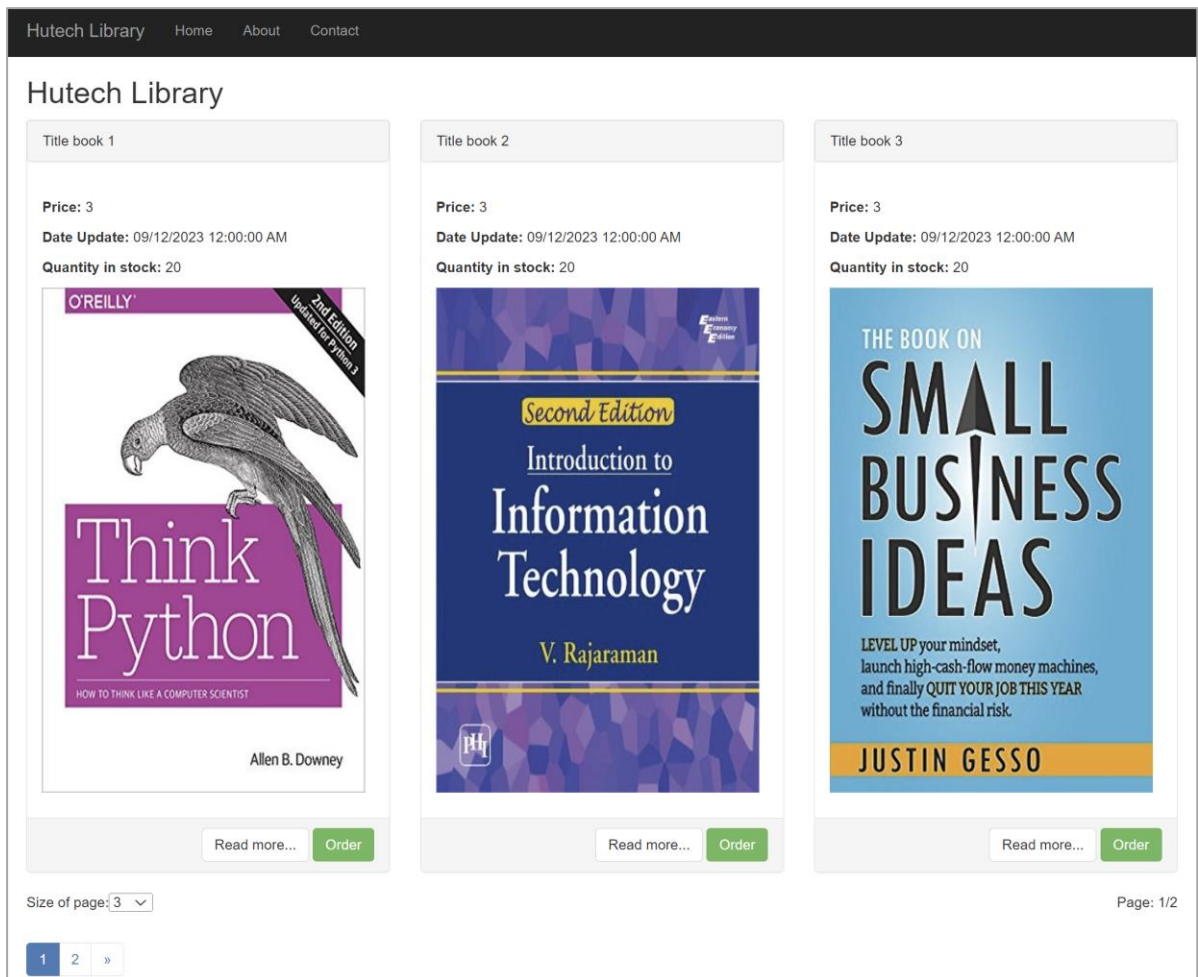
B. LAB 4

- Create a new project or use Lab3 project (ASP.Net MVC Application)

1. Create a connection between a project and database by Linq

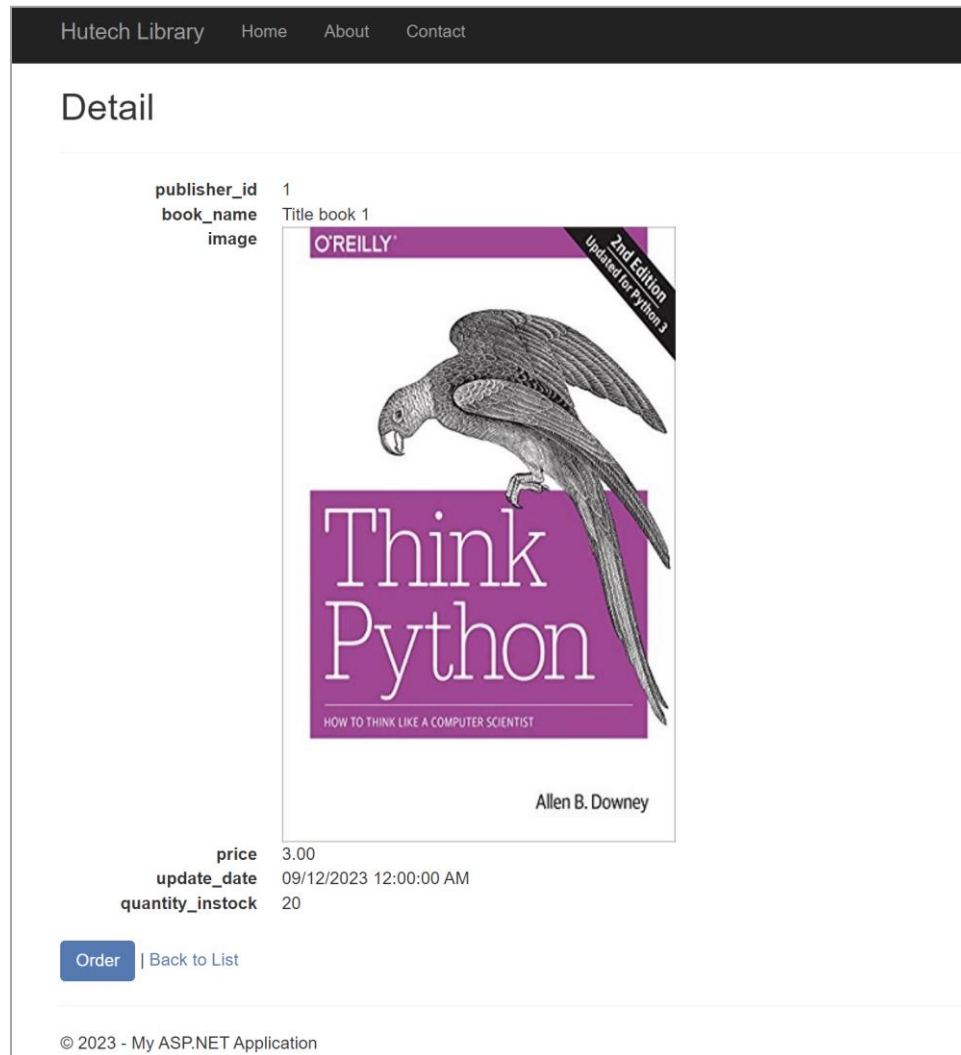


2. Create an Index page



3. Create a Detail page

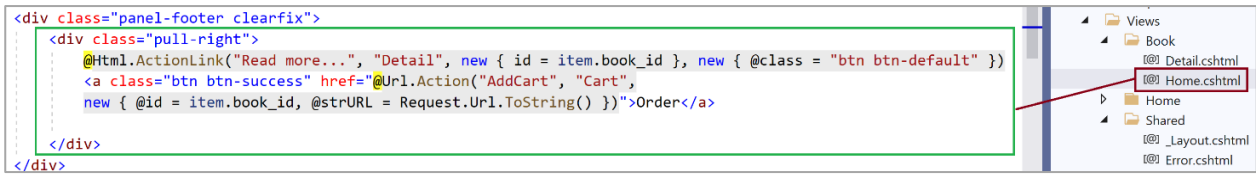
- + Code Detail function in BookController.
- + Right-click the ActionResult Detail → Add View...
- + Layout Detail.cshtml (if possible)
- + Result: **Detail.cshtml** page



4. Create Order function

a. Add menu: Cart()

+ Open Home.cshtml and edit at Order button:



```
<div class="pull-right">
  @Html.ActionLink("Read more...", "Detail", new { id = item.book_id }, new { @class = "btn btn-default" })
  <a class="btn btn-success" href="@Url.Action("AddCart", "Cart", new { @id = item.book_id, @strURL = Request.Url.ToString() })">Order</a>
</div>
```

+ Add Cart menu in _Layout.cshtml file (Views / Shared)



```
<li>@Html.Partial("~/Views/Cart/CartPartial.cshtml")</li>
```

b. Add model: Cart.cs

+ Right-click on Models → Add → New Item ... → Code → **Cart.cs**

```
public class Cart
{
    dbBookStoreDataContext db = new dbBookStoreDataContext();
    public int book_id { get; set; }
    [Display(Name = "Book name")]
    public string book_name { get; set; }

    [Display(Name = "Cover image")]
    public string image { get; set; }

    [Display(Name = "Price")]
    public Double price { get; set; }

    [Display(Name = "Quantity")]
    public int iquantity { get; set; }

    [Display(Name = "Total")]
    public Double Total
    {
        get { return iquantity * price; }
    }
    public Cart(int id)
    {
        book_id = id;
        book b = db.books.Single(n => n.book_id == book_id);
        book_name = b.book_name;
        image = b.image;
        price = double.Parse(b.price.ToString());
        iquantity = 1;
    }
}
```

c. Add controller: *CartController.cs*

+ Right-click Controllers → Add → Controller... → File name: CartController

```
public class CartController : Controller
{
    dbBookStoreDataContext db = new dbBookStoreDataContext();
    public List<Cart> getCart()
    {
        List<Cart> lstCart = Session["Cart"] as List<Cart>;
        if (lstCart == null)
        {
            lstCart = new List<Cart>();
            Session["Cart"] = lstCart;
        }
        return lstCart;
    }
    public ActionResult AddCart(int id, string strURL)
    {
        List<Cart> lstCart = getCart();
        Cart product = lstCart.Find(n => n.book_id == id);
        if (product == null)
        {
            product = new Cart(id);
            lstCart.Add(product);
            return Redirect(strURL);
        }
        else
        {
            product.iquantity++;
            return Redirect(strURL);
        }
    }
    private int sumQuantity()
    {
        int tsl = 0;
        List<Cart> lstCart = Session["Cart"] as List<Cart>;
        if (lstCart != null)
        {
            tsl = lstCart.Sum(n => n.iquantity);
        }
        return tsl;
    }
    private int sumProductQuantity()
    {
        int tsl = 0;
        List<Cart> lstCart = Session["Cart"] as List<Cart>;
        if (lstCart != null)
        {
            tsl = lstCart.Count;
        }
        return tsl;
    }
}
```

```

private double Total()
{
    double tt = 0;
    List<Cart> lstCart = Session["Cart"] as List<Cart>;
    if (lstCart != null)
    {
        tt = lstCart.Sum(n => n.Total);
    }
    return tt;
}

public ActionResult Cart() //create view
{
    List<Cart> lstCart = getCart();
    ViewBag.sumQuantity = sumQuantity();
    ViewBag.Total = Total();
    ViewBag.sumProductQuantity = sumProductQuantity();
    return View(lstCart);
}

public ActionResult CartPartial() //create view
{
    ViewBag.sumQuantity = sumQuantity();
    ViewBag.Total = Total();
    ViewBag.sumProductQuantity = sumProductQuantity();
    return PartialView();
}

public ActionResult CartDelete(int id)
{
    List<Cart> lstCart = getCart();
    Cart product = lstCart.SingleOrDefault(n => n.book_id == id);
    if (product != null)
    {
        lstCart.RemoveAll(n => n.book_id == id);
        return RedirectToAction("Cart");
    }
    return RedirectToAction("Cart");
}

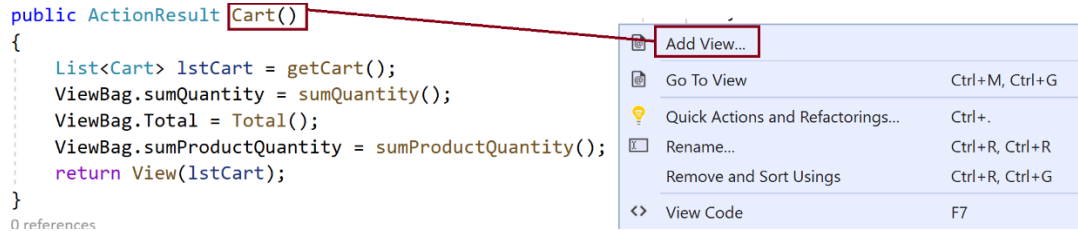
public ActionResult CartUpdate(int id, FormCollection collection)
{
    List<Cart> lstCart = getCart();
    Cart product = lstCart.SingleOrDefault(n => n.book_id == id);
    if (product != null)
    {
        product.iquantity= int.Parse(collection["txtQuantity"].ToString());
    }
    return RedirectToAction("Cart");
}

public ActionResult AllCartDelete()
{
    List<Cart> lstCart = getCart();
    lstCart.Clear();
    return RedirectToAction("Cart");
}
}

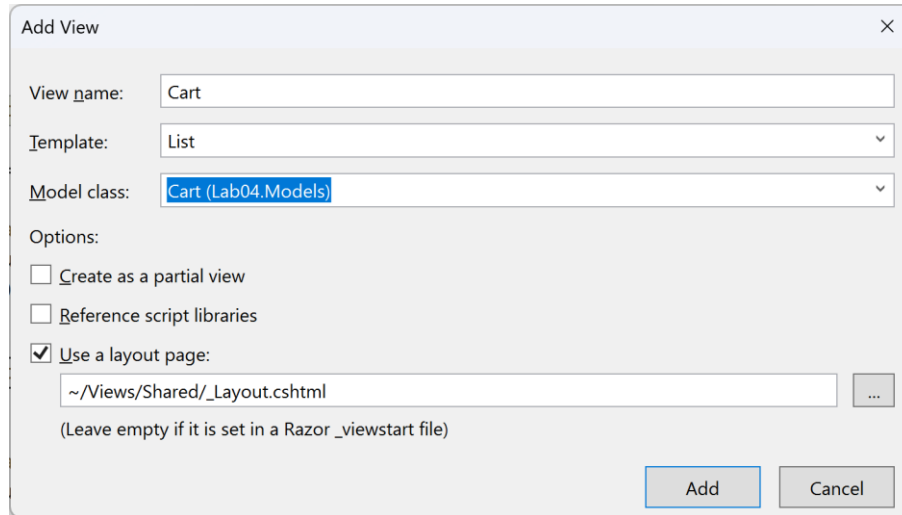
```

d. Add view: *Cart.cshtml*

+ Right-click ActionResult Cart → Add View



+ Create Cart view with Template: List and Model class: Cart



+ Layout code in *Cart.cshtml*

```
@model IEnumerable<Lab04.Models.Cart>  
@{  
    ViewBag.Title = "Cart";  
    Layout = "~/Views/Shared/_Layout.cshtml";  
}  
<h2>Your Cart</h2>  
<table class="table">  
    <tr>  
        <th>@Html.DisplayNameFor(model => model.book_name)</th>  
        <th>@Html.DisplayNameFor(model => model.image)</th>  
        <th>@Html.DisplayNameFor(model => model.price)</th>  
        <th>@Html.DisplayNameFor(model => model.iquantity)</th>  
        <th>@Html.DisplayNameFor(model => model.Total)</th>  
    </tr>  
    @foreach (var item in Model)  
    {  
        <tr>  
            <td> @Html.DisplayFor(modelItem => item.book_name) </td>  
            <td>  </td>  
            <td> @using (Html.BeginForm("CartUpdate", "Cart", new { @id = item.book_id }))  
            {  
                <td> @String.Format("{0:0,0}", item.price) USD </td>  
                <td> <input type="number" min="1" name="txtSolg"  
                    value="@item.iquantity" /> </td>  
                <td> @Html.DisplayFor(modelItem => item.Total) </td>  
                <td>  
                    <a href="@Url.Action("Detail", "Book", new { id=item.book_id })">  
Detail</a> | @Html.ActionLink("Delete", "CartDelete", new { @id = item.book_id })  
                    </td>  
            }  
            </td>  
        </tr>  
    }  
    </tr>  
</table>
```



```

        <input type="submit" value="Update" class="btn btn-success" />
    </td>
    }
</tr>
}
<tr style="font-weight:bold; text-align:left; color:red">
    <td colspan="2"> Quantity Sum: @ViewBag.sumProductQuantity </td>
    <td colspan="2"> Quantity Total: @ViewBag.sumQuantity </td>
    <td colspan="4"> Total: @String.Format("{0:0,0}", @ViewBag.Total) USD </td>
</tr>
<tr style="font-weight:bold; color:blue; text-align:right ">
    <td colspan="5">
        @Html.ActionLink("Cart Delete", "AllCartDelete")
    </td>
</tr>
<tr style="font-weight:bold; color:yellow; text-align:right ">
    <td colspan="5">
        @Html.ActionLink("Place order", "PlaceOrder", "Cart")
    </td>
</tr>
</table>

```

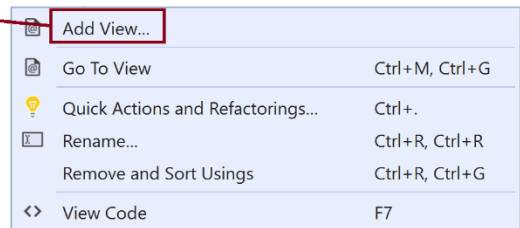
e. Add View: CartPartial.cshtml

+ Right-click on ActionResult CartPartial → Add View

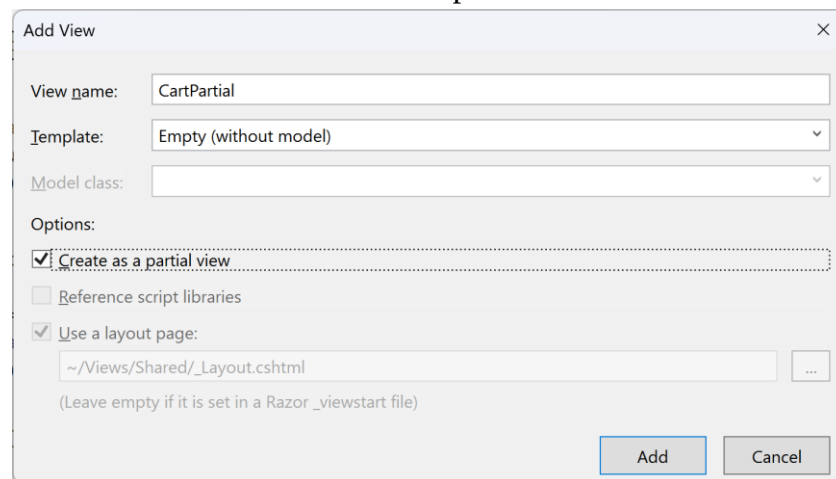
```

public ActionResult CartPartial()
{
    ViewBag.sumQuantity = sumQuantity();
    ViewBag.Total = Total();
    ViewBag.sumProductQuantity = sumProductQuantity();
    return PartialView();
}

```



+ Create CartPartial view with Partial view option:



+ Layout code in **CartPartial.cshtml**

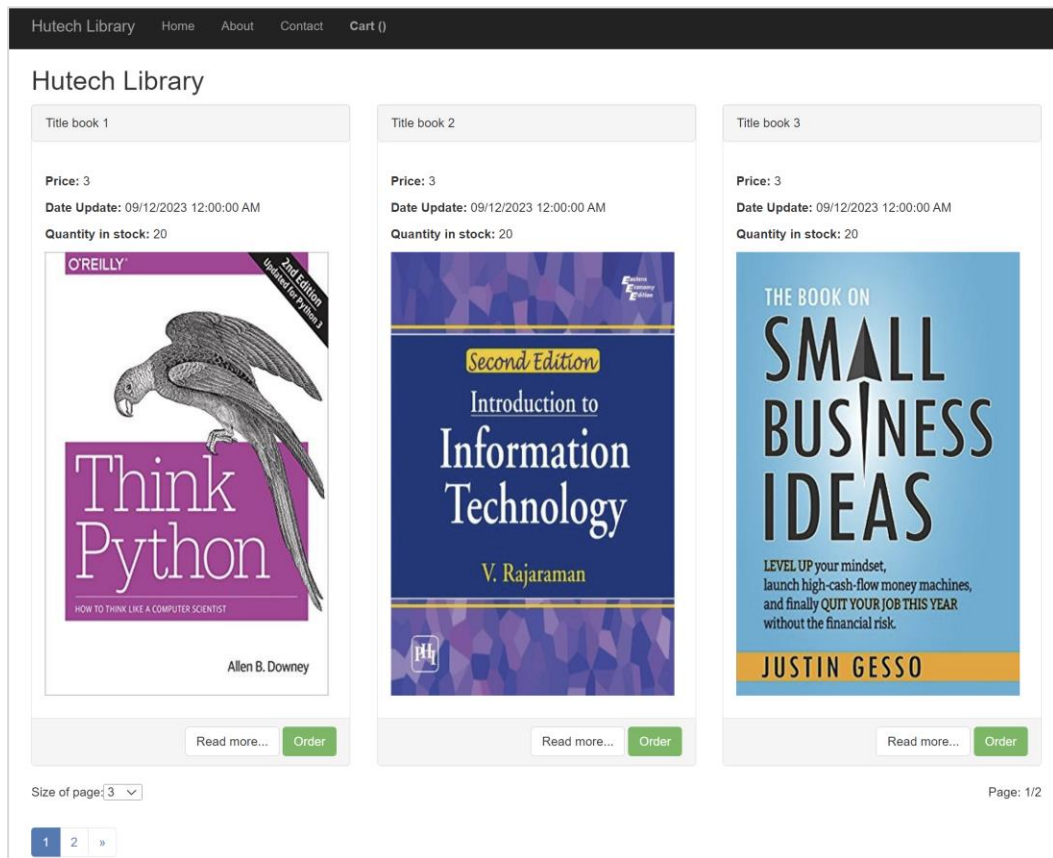
```

<a class="getstarted scrollto active" href="@Url.Action("Cart","Cart")"
style="font-weight:bold; text-decoration:none">
    Cart (@ViewBag.sumProductQuantity)
</a>

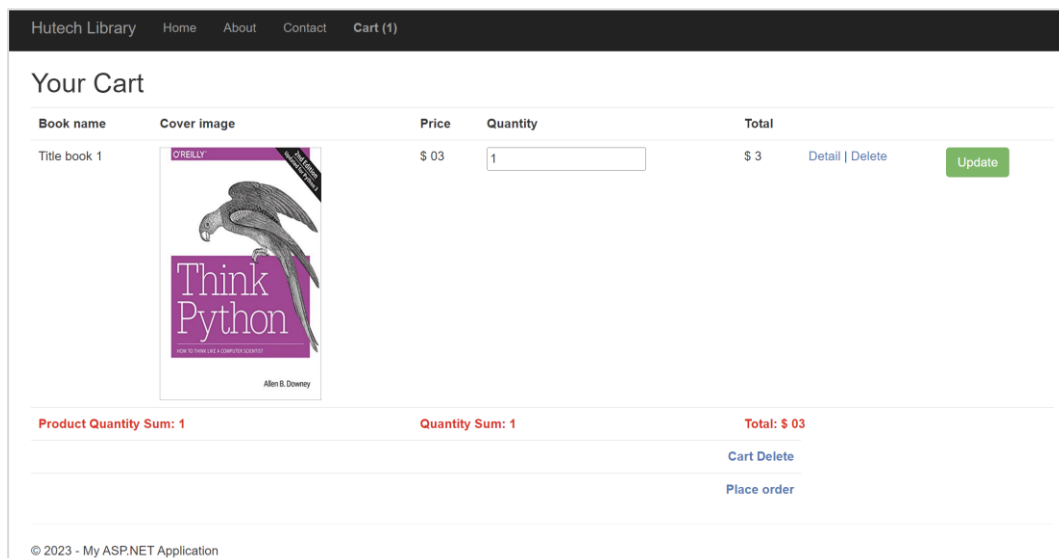
```

+ Result:

→ Home page:



→ Click Order button and open your cart:



→ Check Update, Cart Delete function

5. Build "Place order" function

- Selecting "Place Order" will notify the user "Notification of successful order"
- Save shopping cart information into the database.